Vladimir Vladimirovich Baryshev

Neural decoding | Neurotechnology | Biomedical AI | Computer vision

Email | LinkedIn | GitHub | ORCID

PROFILE

Neurotechnology MSc student and Research Collaborator focused on neural decoding with machine learning. Experience spans intracranial ECoG speech decoding, EEG classification, EMG/IMU gait-phase decoding, computer vision, and biomedical image analysis, with hands-on work across research design, data processing, neural-network training, and scientific writing.

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Collaborator - HSE University, Centre for Bioelectric Interfaces

Part-time, remote | Moscow, Russia | Jun 2026 - Present

Develop intracranial ECoG speech-decoding methods. Best internal experiment: 81.4% accuracy, +11.4 percentage points over the previous 70.0% benchmark. All other methodological and dataset details remain under NDA.

Independent ML Developer / Researcher

Self-employed | Apr 2026 - Present

Develop independent ML research in pelvic CT segmentation and EEG decoding; the EEGNet JapanDataset reproduction is currently on hold after a reproducibility audit.

Research Assistant / Computer Vision Intern - Skolkovo Institute of Science and Technology, Neurocenter

Research Assistant, part-time: Nov 2025 - Apr 2026 | Intern: Sep 2025 - Nov 2025

As Research Assistant, built an invasive myodecoder integrating six-channel EMG, thigh/shank IMUs, YOLO11s-Pose, and Unity 3D; CNN-BiLSTM PhaseNet reached 94.3% internal window-level validation accuracy. As Intern, built a real-time sheep pose-estimation pipeline reaching 28.9 FPS at 640 px FP16 on a laptop RTX 5070, with Kalman-smoothed keypoints.

EDUCATION

MSc, Neurotechnology with the Basics of Biomaterials Science

Sirius University, Sochi | In progress

Master's thesis: spatiotemporal neuromodulation and biomimetic neurointerfaces. Developing an adaptive closed-loop neuromodulation approach for personalized neurorehabilitation.

BSc, General Biology - Molecular Biology and Genetics

North-Caucasus Federal University, Stavropol | GPA 4.35

Developed software for morphological processing of histological slice images and object segmentation; work supported by Russian Science Foundation project 23-71-10013.

PUBLICATION

Lyakhov P. A., Lyakhova U. A., Baboshina V. A., Baryshev V. V., Nagornov N. N. Detection of attention state in children with autism spectrum disorder based on neural network classification of electroencephalograms. Vestnik of Saint Petersburg University. Applied Mathematics. Computer Science. Control Processes, 2025, 21(1), 92-111. Q3 Scopus. Four-MLP weighted-average ensemble: 95.90% accuracy, F1 0.9590, MCC 0.9183; +2.92 pp over the prior reported best on a single-participant dataset. DOI

SELECTED PROJECTS

Attention-state detection in children with autism spectrum disorder

EEG classification | Published

Led the research cycle and co-authored the paper. A four-MLP weighted-average ensemble reached 95.90% accuracy, F1 0.9590, and MCC 0.9183 on 33,936 balanced samples from one participant - 2.92 pp above the prior reported best.

Invasive Myodecoder - EMG/IMU gait-phase decoding

Multimodal neural decoding | Completed

Built a six-channel invasive EMG and thigh/shank IMU pipeline with YOLO11s-Pose and Unity 3D. CNN-BiLSTM PhaseNet classified stance and swing with 94.3% internal window-level validation accuracy; predictions streamed over TCP. GitHub

Real-Time Sheep Pose Estimation with YOLO11-Pose

Computer vision | Completed

Built an FP16 real-time animal pose pipeline with Kalman-smoothed keypoints, reaching 28.9 FPS at 640 px on a laptop RTX 5070.

Pelvic CT segmentation support system

3D medical imaging | In progress

Build an ML-based clinical decision-support prototype for segmentation of pelvic bones and lower limbs. GitHub

EEGNet classifier on JapanDataset

EEG deep learning | On hold

Audited 10 sessions of 128-channel EEG. Balanced accuracy: 55.6% / 54.9% / 33.4% for overt / minimally overt / covert speech versus 94.6% / 94.9% / 91.1% reported; identified likely checkpoint and train/evaluation preprocessing inconsistencies whose causal contribution remains unconfirmed. GitHub

Histological slice processing software

Python desktop application | Completed

Built an end-to-end image-processing application for color correction, channel normalization, threshold binarization, and morphological operations using OpenCV, NumPy, Matplotlib, and PyQt.

TECHNICAL SKILLS

- Programming: Python; basic Git and Linux
- ML and scientific computing: PyTorch, NumPy, MNE, Matplotlib
- Neural and biomedical data: ECoG, EEG, EMG, IMU, MRI, CT, neural decoding, signal preprocessing
- Computer vision: OpenCV, YOLO11-Pose, 3D U-Net, nnU-Net, SAM, DeepLabCut
- Application development: Unity, TCP integration, PyQt

SELECTED TRAINING

- Neuromatch Academy - Computational Neuroscience, 2025
- New Generation Neurointerfaces, HSE University, 2025
- HSE Brain Research School, 2024
- Computational Neuroscience, University of Washington (Coursera)
- Neuroscience and Neuroimaging Specialization, Johns Hopkins University (Coursera)

SCIENCE COMMUNICATION AND LANGUAGES

Staff science writer at Science Mail with more than 800 popular-science articles across neuroscience, genetics, physics, artificial intelligence, and biomedicine.

- Russian: native
- English: B2 (intermediate-upper intermediate)